

RESEARCH ARTICLE

Simul-RL Portfolio Framework: Black-Scholes-Merton and Reinforcement Learning for Asset Allocation

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ABSTRACT Asset allocation method using reinforcement learning is being actively researched. However, the existing asset allocation methods do not consider the following viewpoints in solving the asset allocation problem. First, State design without considering portfolio management and financial market characteristics. Second, Model Overfitting. Third, Model training design without considering the statistical structure of financial time series data. To solve these problems, we propose a new Reinforcement Learning asset allocation method. First, financial market state and agent state. Second, Monte Carlo simulation data are used to increase training data complexity. Third, Monte Carlo simulation data are created considering various statistical structures of financial markets. We show experimentally that our method outperforms the benchmark at several test intervals.

INDEX TERMS Asset allocation, Black-Scholes-Merton, finance, reinforcement learning, simulation data.

I. INTRODUCTION

Investment refers to allocating time and money in anticipation of future profits [1]. There are many different ways to allocate time and money. For example, there is a method of depositing an investment in a bank, a method of investing in a stock-type fund, a method of investing in a bond-type fund. These methods are either low- or high-returns methods; however, they are prone to market risks and may result in a loss of principal. It is important to adjust the ratio of risky and safe assets to avoid low-return or high-risk situations. A portfolio developed with these assets should be managed with low risk and high expected return. Asset allocation allocates money to different assets with different expected returns and different levels of risk. Thus, a portfolio is developed for achieving low volatility and long-term stable returns [1], [2]. As a method of asset allocation, there is a traditional asset allocation method, and recently, an asset allocation method using machine learning is being developed.

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There are many different method for traditional asset allocation, such as Markowitz and Risk Budgeting. In Markowitz, the historical prices of assets are used to derive efficient frontier, which are subsequently used to derive an optimal portfolio [3]. Risk budgeting estimates portfolio weights by assigning a portfolio's risk to each asset [4]. Recently, financial AI techniques using machine learning are being actively studied. The results of these studies are being used in areas such as market forecasting, asset allocation, risk management, and market timing [5], [6], [7], [8]. Asset allocation using Reinforcement Learning has more flexibility in setting goals and using a variety of input variables compared to the existing asset allocation model. Such a variety of information includes benchmark management information, portfolio management information, market returns, market price prediction information, market volatility information, basic information of stocks, macro variables.

However, The existing asset allocation methods do not consider the following viewpoints in solving the asset allocation problem.

A. PROBLEM 1: STATE DESIGN

In general, asset allocation method using reinforcement learning use the return of assets as the state. This state is related with the environment. That is, it is considered as a financial market state. Financial market states contain a lot of noise, and it is difficult for the financial market state to have a state transition that is affected by the action of an agent. If the state affected by the action does not exist in the agent, the agent cannot recognize what state it is currently in. This can cause a very big problem in that the agent decides an action without recognizing its current situation. Therefore, we define the status of the portfolio managed by the agent as the agent state, and use it as an input to the agent along with the financial market state.

B. PROBLEM 2: MODEL OVERFITTING

Assuming that financial time series daily data for 20 years is used to train the model, one stock or ETF has approximately 4800 daily data point. Machine learning method use historical financial time series data as training, validation and test. In this data split, machine learning architectures can have high model complexity compared to data complexity. Therefore, the probability that the model overfits the training data is very high [9], [10]. An overfitted model can perfectly manage the portfolio for training data, but the model does not manage the portfolio well for validation and test data. When performing asset management on real data using an overfitted model, a problem in which the output of the model cannot be trusted may occur. This is a very very serious problem from the perspective of asset management that manages customers' money. Therefore, a method to prevent model overfitting is very important in the field of finance using artificial intelligence.

C. PROBLEM 3: REGIME CHANGES (STATISTICAL STRUCTURE CHANGE)

Because financial time series data has non-stationary characteristics, the statistical structure of financial time series data is constantly changing [5], [11]. However, the existing asset allocation methods using machine learning do not consider the changes in the statistical structure of financial time series data in model learning and inference. A model trained on only a specific statistical structure of financial time series data is difficult to cope with problems that may arise when the statistical structure of the market changes. And the asset allocation weight calculated by this model is not an optimized value for the changed statistical structure. From an asset management perspective, if the model does not produce an optimal asset allocation weight, it can result in very serious losses. To alleviate this problem, it is very important to consider the various statistical structures of the market in training the model.

To solve the above problems, we propose a new Reinforcement Learning asset allocation method. First, the state of the portfolio managed by the model is considered as

the state of the agent, and the daily return vector of the asset is considered as the state of Financial market. Second, we increase the complexity of the training data using Monte Carlo simulation data to prevent model overfitting. This can alleviate the problem of model overfitting. Third, various statistical structures are defined for financial markets. Then, simulation data is generated for these statistical structures and the model is trained.

The contributions of this paper are as follows. First, we use Monte Carlo simulation to mitigate model overfitting. Second, by using domain generalization, this paper mitigates the non-stationary feature that the statistics of the random process that generates financial time series continuously change. Through them, this paper solves the unstable problem of the model in unseen data. This paper proposes an AI asset allocation framework that is very suitable for rapidly changing financial markets with little learning data. This paper is organized as follows. Section II describes related work, Section III describes motivation, Section IV describes proposed method, Section V present the experiments, Section VI describes conclusion and future work.

II. RELATED WORK

Traditional asset allocation method are still being actively studied and applied to finance. Reference [12] proposed a new probabilistic risk budgeting multi-portfolio optimization. This model performs an integrated optimization of the individual security responsible for marginal risk contribution. Reference [13] applied the Markowitz methodology to develop a new strategy algorithm for long-term stocks. This methodology optimizes portfolios based on high potential returns and high risk strategies.

There are various method for asset allocation using Reinforcement Learning. Reference [14] applied a simple Deep Reinforcement Learning method to US Equity. Reference [15] built an asset allocation model using a GRU network and a Deep Reinforcement Learning (DRL) algorithm. This model uses previous state, action to extract the current action. And the model was trained using risk-adjusted returns. Reference [16] used the S&P 500 and U.S. Treasury bonds as assets, used adaptive learning, and built a retirement pension trading portfolio model. This paper uses the prices of the S&P 500 and bonds as the agent's state. Reference [17] used financial article data and stock price data to generate stock movement signals, and used the signals and stock price data to build an asset allocation Reinforcement Learning agent. This paper uses stock price and news data as an augmented state. Reference [18] presents a comprehensive description of the efficiency and advantages of modeling and representing a trading agent based on traditional system identification and model-free agent-based methods. Reference [18], observation and state are used as inputs to the reinforcement learning agent. observation is the price of an asset. As the state, the asset allocation ratio and the cumulative rate of return of assets are used. Reference [19]

developed a portfolio Reinforcement Learning agent using three subnetworks. The first subnetwork is a network that handles returns and volatility of assets, and the second subnetwork is a network that handles contextual information such as correlations between assets. The third subnetwork handles the previous portfolio weights. Above Studies that solve asset allocation problems using deep reinforcement learning have used asset prices, financial market news, cumulative returns on assets, and asset allocation proportions as states or observations. It is difficult to think that the information of the portfolio being managed can be changed by the action of the agent.

Reference [20] created a market simulation for the base index by utilizing GAN in a mathematical programming framework and built a portfolio. And this methodology showed that the portfolio works well out of sample. Reference [21] formulated the portfolio optimization problem as an asset allocation problem accessing each asset class through passive (index) funds. The optimizer integrates the context information and uses the Modified CTGAN algorithm to generate synthetic return scenarios, which are fed into the optimization engine. Reference [22] introduces a deep generative model-based autoencoding CGAN (ACGAN) that learns internal trends from past data while modeling market uncertainty and future trends. This methodology shows that the proposed ACGAN model leads to better portfolio allocation and produces series that are closer to real data compared to the existing Markowitz and CGAN approaches. Above studies created simulation data using GAN, and used it to solve various financial market problems including asset allocation. However, GAN [23] are unstable in network learning. In addition, the non-stationary characteristics of financial markets were not considered in generating simulation data. It models variously changing financial markets with simple statistical parameters.

III. MOTIVATION

A. DATA AUGMENTATION IN FINANCIAL MARKET

Problems that may occur in financial time series data, where learning data is very sparse, can be solved with simulation data. In the AI field, GAN [23] that can generate simulation data is being actively researched. GAN is also widely used in finance domain where learning data is very sparse [24], [25], [26]. However, GAN has the disadvantage of unstable neural network learning. A big problem can arise if a model that learns the output of a GAN is used for asset management. So, We use monte carlo simulation as a way to generate simulation data.

Monte Carlo simulation is a numerical method that can be effectively applied to problems with unknown or complex closed solutions [27]. Monte Carlo simulations are used in the valuation of derivatives. The stochastic process mainly used for the movement of financial assets is the geometric Brownian motion. A stochastic process S_t is said to follow geometric Brownian motion when it satisfies the following

stochastic differential equation:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1)$$

In this case, μ is the return of the underlying asset, σ is the volatility of the return, and W_t is the winner process. When the initial value is S_0 , the solution S_t in the above equation satisfies the following equation.

$$S_t = S_0 \exp[(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t] \quad (2)$$

We can also derive the following equation to run the Monte Carlo simulation. Δt is the time interval and ϵ_t is a standard normal random number. Using the above formula, we can create multiple stock paths. And we use multiple stock paths to calculate the payoff for the option and determine the option price.

$$S_{t+\Delta t} = S_t \exp[(\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma \epsilon_t \sqrt{\Delta t}] \quad (3)$$

In addition, the stock price path of assets can be created using correlation of assets and Monte Carlo simulation. These stock paths satisfy a specific correlation between assets. And thousands or tens of thousands of asset paths can be created through a specific correlation matrix. And these asset paths have various patterns that can be expressed in real financial markets. There are many ways to solve the model overfitting problem through simulation data generation, but our method is the first to stably solve model overfitting using monte carlo simulation.

B. REGIME CHANGES IN NON-STATIONARY FINANCIAL TIME-SERIES

Financial time series data are generally non-stationary [11], [28]. This means that the mean and variance of the random process that generates financial time series data are constantly changing.

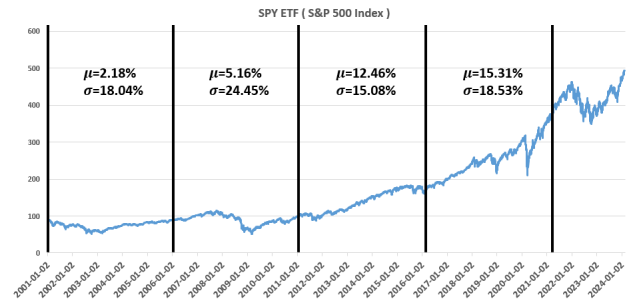


FIGURE 1. Financial time series data has a non-stationary characteristic. This means that the statistics of financial time series data change as time changes. In the figure above, the statistical values (mean, standard deviation) of the data return rate for each section continue to change. This means that the parameter values of the distribution that generates the data change. In other words, the parameters of the 'distribution that generates the data for financial time series data change as time changes.

If the statistics of the random process that generate financial time series data change, the model should be optimized for the changed statistics. However, this characteristic

is not considered in the existing asset allocation method using machine learning. It simply learns only the statistical structure of the financial market included in the training data.

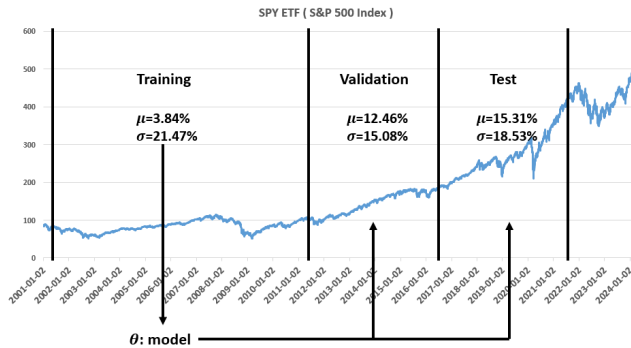


FIGURE 2. The figure above shows the model learning and evaluation methods commonly used in machine learning. We can see that the distribution of financial time series data in the training period is different from the distribution of financial time series data in the validation and test periods. The learning/evaluation methods commonly used in machine learning do not reflect the non-stationary characteristics of financial time series data. In other words, the model learned with data in the training period may not work well in the validation and test periods. Therefore, when learning a machine learning model for financial time series data, various regimes should be considered.

In order to alleviate the non-stationary disadvantage of financial time series data, we divide financial time series in various regimes and utilize them. Here, each regime is defined as a correlation matrix of assets constituting the financial market. For model training, in each regime, we generate a lot of simulation data using correlation matrix and Black Scholes Merton. And we use this data to learn a model and define a model set. For each regime, one model set is produced. For model inference, we find the regime most similar to the correlation matrix computed at time t . And we bring the models learned in the regime and make inferences. This method is similar to domain generalization [29] in meta learning. In other words, we alleviate the non-stationarity of financial time series data through generalization.

IV. PROPOSED METHOD

A. SIMUL-RL PORTFOLIO FRAMEWORK

The Simul-RL Portfolio Framework works as follows. The framework loads the original price data of stocks/bonds/commodities. Then, it calculates the returns using the price data. It performs the RCME Process using the returns and extracts regimes. It generates a simulation price data set using the extracted regimes and the Black-Scholes-Merton method. After that, it performs the Model Learning Process to train the models and generates the total model pool. The data used to train the model is a simulation data set. Then, it performs the Model Inference process using the Unseen data and the Total Model Pool and calculates the asset allocation ratio.

The components of the Simul-RL Portfolio Framework are as follows. First, the Data Analysis Part extracts regimes and generates simulation data set. Second, the Model Part trains models using the simulation data set and infers the asset

allocation weight. We explain each component in detail in the following sections.

B. DATA ANALYSIS PART

1) CORRELATION MATRIX EXTRACTION PROCESS

The statistical structure of the financial market is considered as a correlation matrix of the assets composing the financial market. We defined the statistical structure of financial markets as representative correlation matrix set.

To extract the representative correlation matrix set, we define the Representative Correlation Matrix Extraction Process (RCME Process). The RCME process divides financial time series data into several specific financial time series intervals and extracts a representative correlation matrix from the intervals. Each representative correlation matrix extracted from each interval is different. The detailed RCME Process is as follows. At each time t , the correlation matrix is calculated using the daily closing prices of the assets over the past year. And the process creates a correlation matrix set (CMS) using these correlation matrices. The size of the CMS is T because the correlation matrix is computed at every time of the financial time series data, where T is the total length of the financial time series data daily closing price. Process calculates Correlation Matrix Distance Matrix (CMDM) using CMS. Each element of CMDM is defined as follows.

Definition (CMDM Element $CMDM_{i,j}$):

$$CMDM_{i,j} = \|A^{i,j}\|_F, i, j = 1, \dots, T \quad (4)$$

Definition (Frobenious norm with two correlation matrix):

$$\|A^{i,j}\|_F = \sqrt{\sum_{k=1}^n \sum_{l=1}^n |a_{k,l}|}, \quad (5)$$

where $A^{i,j} = \Sigma_i - \Sigma_j$, $\Sigma_i \in CMS$, $\Sigma_j \in CMS$

After the CMDM is created, the Process executes Hierarchical Clustering [30] to create a cluster set (CS).

For each cluster generated by hierarchical clustering, the process computes a representative correlation matrix. The representative correlation matrix is calculated as the average matrix of the correlation matrices belonging to the cluster.

Definition (Representative Correlation Matrix):

Representative correlation matrix for a specific cluster

$$\bar{\Sigma}^i = \frac{1}{k} \sum_j^k \Sigma_j^i \quad (6)$$

where Σ_j^i : j -th correlation matrix in i -th cluster

The Representative correlation matrix represents the cluster. This matrix can be considered to be a correlation matrix representing a particular interval of financial time-series data.

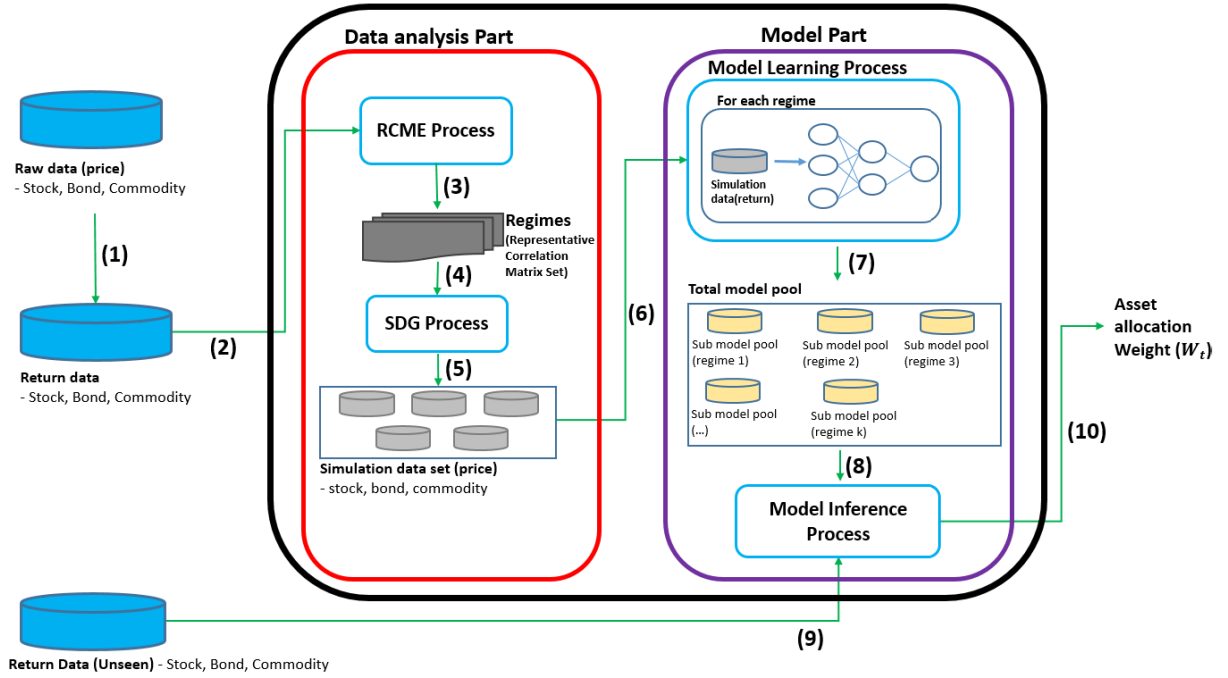


FIGURE 3. The figure above shows the model learning and evaluation methods commonly used in machine learning. We can see that the distribution of financial time series data in the training period is different from the distribution of financial time series data in the validation and test periods. The learning/evaluation methods commonly used in machine learning do not reflect the non-stationary characteristics of financial time series data. In other words, the model learned with data in the training period may not work well in the validation and test periods. Therefore, when learning a machine learning model for financial time series data, various regimes should be considered.

Algorithm 1 RCME Process

Input: Financial Time Series Data Set
Output: Representative Correlation Matrix Set

```

for  $i \leftarrow 1$  to  $T$  do
     $\Sigma_i \leftarrow$  calculate correlation matrix at time  $i$ 
     $CMS \leftarrow \Sigma_i$ 
end
for  $i \leftarrow 1$  to  $T$  do
    for  $j \leftarrow 1$  to  $T$  do
         $CMDM_{i,j} \leftarrow \|A^{i,j}\|_F$ 
    end
end
CS  $\leftarrow$  Hierarchical Clustering(CMDM)
for  $i \leftarrow 1$  to  $K$  do
     $\bar{\Sigma}^i \leftarrow$  average correlation matrices in  $i$ -th cluster
end

```

2) SIMULATION DATA GENERATION PROCESS(SDG PROCESS)

We generate financial time series simulation data using the representative correlation matrix calculated by the RCME process. We use the Black-Scholes-Merton method to generate the financial time series simulation data [31]. The length of the financial time series simulation data is 10 years, which is the closing price. The financial time series simulation data generated by Black-Scholes-Merton [31]

are various simulation data corresponding to the correlation matrix, the statistical structure of the financial market. The data includes various patterns. The actual financial time series data is only one realization path data realized by the correlation matrix, and only specific patterns are included in the data. In machine learning methods using financial time series data, these various simulation patterns can increase data complexity and prevent model overfitting.

C. MODEL PART

1) STATE, REWARD

Financial Market State is defined as a return matrix consisting of past 60 day return vectors based on time t for individual assets. The definition of the return vector and return matrix is as follows.

Definition (Financial Market State):

$$\bar{r}_t = \begin{bmatrix} r_t^1 \\ \vdots \\ r_t^k \end{bmatrix} \quad (7)$$

where: r_t^i is the historical return vector of the i th asset at time t

The state of an agent is defined as the return vector for past 120 days of the portfolio managed by the agent at time t .

Definition (Agent State):

$$pr_t = (pr_{t-119}, pr_{t-118}, \dots, pr_t) \quad (8)$$

where: $r_t = \frac{\text{portfolio value}_t}{\text{portfolio value}_{t-1}} - 1$

In general, Reinforcement Learning agents in portfolio management tend to use asset daily return as the agent's state. However, this is not information based on the agent's actions. This is simply information about the market. Therefore, it is reasonable to define the return of the portfolio depending on the agent's action as the agent's state. The reward is defined as the sharpe ratio of portfolio daily return [32].

Definition (Sharpe Ratio): -Reward, objective function

$$\text{SharpeRatio} = \frac{\mu_{\text{port}}}{\sigma_{\text{port}}} \quad (9)$$

μ_{port} and σ_{port} is the average value and the volatility of the portfolio annualized returns managed by the agent.

2) MODEL ARCHITECTURE

As the input of the Reinforcement Learning model, the Historical Return Matrix is used for Financial Market State, and the Historical Portfolio Management Return is used for Agent State. The features for Financial Market State are extracted by applying convolution layers to observation. And the features for Agent state are extracted by applying convolution layers to the state. And the nonlinear relationship is modeled by applying the features extracted from states to the FC layer. The result of FC Layer is used as input for value function and policy function. The architecture is as follows.

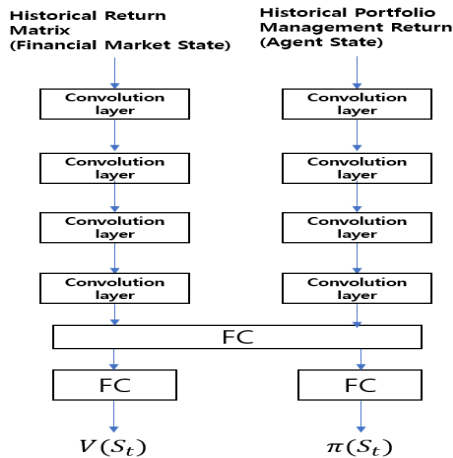


FIGURE 4. The above figure shows the model architecture. The input of the model is divided into financial market state and agent state. A convolution filter is applied to each input and mapped to a fully connected layer. Then, the value and policy values are calculated.

A3C [33] is used as the architecture learning algorithm. A3C is a reinforcement learning algorithm that trains a deep neural network using a gradient descent algorithm asynchronously. Details on the parameters of the neural network architecture are described in table 1.

3) MODEL LEARNING PROCESS

The learning process selects one of the correlation matrices generated by the RCME process. A simulation data set is created using the correlation matrix and the SDG process. a simulation data included in the simulation data set is the

TABLE 1. Neural network architecture parameters.

Parameters	values
# of Convolution layers (for historical return)	4
# of Convolution layers (for portfolio return)	4
# of filters (for each convolution layer)	10
# of FC layers	1,1(value),1(policy)
Activation Function	ReLU
Learning Rate	0.0001
Batch Size	32
Optimization Algorithm	Adam, A3C

simulation data of assets for Equity, Bond, and Commodity. The agent neural network is trained using a simulation data and the A3C algorithm [33]. The inputs of the agent neural network are financial market state and agent state using a simulation data. And the reward of the agent neural network is the sharpe ratio using a simulation data. An agent neural network trained on a simulation data is stored in a sub model pool. The sub model pool containing the agent neural networks trained on the simulation data set is stored in the total model pool. The above process is performed for all correlation matrices belonging to RCMS, and the total model pool is completed.

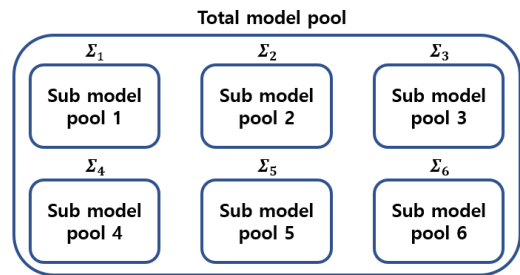


FIGURE 5. Each sub model pool is matched to a specific regime. That is, simulation data is generated using a specific regime (correlation matrix), and models are trained using the data. And the set of trained models is called a sub model pool. And the total model pool is the set of these sub model pools. In other words, it represents the set of models for all regimes.

Algorithm 2 Model Learning Process

Input: RCMS - Representative Correlation Matrix Set

Output: TMP - Total Model Pool

for each Σ **in** RCMS **do**

 Initialize Sub Model Pool

 Simulation Data Set(SDS) $\leftarrow$ SDG process(Σ)

for each simulation data in SDS **do**

 Sub Model Pool $\leftarrow$ A3C(simulation data)

end

 TMP $\leftarrow$ Sub Model Pool

end

4) MODEL INFERENCE PROCESS

Model inference is performed using real data, not simulation data. Model inference is performed as follows. The inference

process calculates the correlation matrix using the past 100 daily closing prices at time t . The inference process calculates the similarity between the correlation matrix and representative correlation matrices, and selects the most similar representative correlation matrix. The inference process calculates asset allocation weights at time t using models of the sub-model pool related to the representative correlation matrix, and calculates the final asset allocation weight by averaging them.

Algorithm 3 Model Inference Process at Time t

Input: RCMS - Representative Correlation Matrix Set,
AAWS - Asset Allocation Weight Set

Output: Asset Allocation Weight $_t$

Initialize AAWS

$\Sigma_t \leftarrow$ Calculate correlation matrix at time t

Input $_t \leftarrow$ Calculate model input at time t

$i \leftarrow \text{argmin}_i$

$\{\|A^{t,i}\|_F | A^{t,i} = \Sigma_i - \Sigma_t, \Sigma_i \in \text{RCMS}\}$

for each model in i -th Sub Model Pool do
| AAWS $\leftarrow$ model(Input $_t$)

end

Asset Allocation Weight $_t \leftarrow$ Average(AAWS)

V. EXPERIMENT

A. CHARACTERISTICS OF EXPERIMENT DATA

Experimental data can be divided into three types: stocks, bonds, and commodity.

bond data is considered as a means of risk avoidance for stocks from the perspective of asset management. In particular, stocks tend to fall while bonds tend to rise in price during risky market situations. In addition, bonds generally have lower returns than stocks and have lower volatility, so they are preferred as safe assets.

stock data is recognized as a major means of obtaining returns in asset management and has the characteristics of high volatility and high returns. These characteristics show that stocks are suitable for more aggressive investment strategies.

commodity is a hedge method against inflation. In an inflation situation, the price of commodity tends to rise. In other words, commodity are used as a means of obtaining returns in an inflation situation.

The characteristics of each asset can be explained by the following statistics. In order to calculate the statistics, representative indices for each asset are set, and the statistics are calculated using the corresponding indices.

The MSCI ACWI Index [34] is a representative index for world country stocks. The Bloomberg Global Aggregate Index [35] is a representative index for world bonds. The S&P GSCI Index [36] is a representative index for commodities.

TABLE 2. Representative indices and statistics for stock, bond, and commodity.

Index Name	Return	Volatility
MSCI ACWI Index	9.26 %	18.76 %
Bloomberg Global Aggregate Index	-0.95 %	4.14 %
S&P GSCI Index	5.21 %	22.49 %

B. EXPERIMENT DATA

Exchange-traded funds (ETFs) are used to experiment with global asset allocation, bond asset allocation, and sector asset allocation. The experimental datasets, namely ALL, DM, Global Bond and US Sector, are built using these ETFs. These experimental data sets exhibit different and distinct characteristics. By conducting experiments on multiple rather than a few datasets, we show that our method outperforms benchmarks. Details of the experimental data are described in Tables 2 and 3.

C. BENCHMARK

To evaluate the proposed framework, we use traditional asset allocation models and equal weight models as benchmarks. The Markowitz(MV) and Risk Budgeting(RB) model is used as a traditional asset allocation model. The risk budgeting model uses a risk parity model. Equal weight(EW) model allocates the ratio of asset allocation to each investment asset as $1/n$. In addition to the traditional model, we add deep learning-based reinforcement learning models as benchmarks(DQN, A3C).

D. EXPERIMENT METRIC

We set annualized return, annualized volatility [37], Sharpe ratio [32], MDD [38], Calmar Ratio [39], [40], Utility Function [1], [41], [42] as statistics for evaluation.

Definition (Annualized Return(R)):

$$r_y = \bar{r} \times 252 \quad (10)$$

where: $\bar{r} = \frac{1}{n} \sum_i^n r_i$, r_i is the daily return of the portfolio. n is the portfolio management period.

Definition (Annualized Volatility(Sigma)):

$$\sigma_y = \bar{\sigma} \times \sqrt{252} \quad (11)$$

where: $\bar{\sigma} = \frac{1}{n-1} \sum_i^n (r_i - \bar{r})^2$

Definition (Sharpe Ratio):

$$sharpe = r_y / \sigma_y \quad (12)$$

MDD [38] refers to the maximum decline from the previous high during portfolio management. calmar ratio is the return divided by the MDD [39], [40].

We want to apply the perspective of “real investment” to the experiment. This is not simply to provide statistics. It is to compare methodologies that can be used in “real asset management”. In other words, we identify methodology that can be “actually” utilized in asset management. This is really

TABLE 3. Exchange Traded Fund(ETF) list.

Ticker	ETF Name	Type
(1) VOO US Equity	Vanguard S&P 500 ETF	Equity(US)
(2) ISF LN Equity	iShares Core FTSE 100 UCITS ETF	Equity(UK)
(3) XDAX GR Equity	Xtrackers DAX UCITS ETF	Equity(DE)
(4) 1329 JP Equity	iShares Core Nikkei 225 ETF	Equity(JP)
(5) FXI US Equity	iShares China Large-Cap ETF	Equity(CH)
(6) 277630 KS Equity	Mirae Asset TIGER KOSPI ETF	Equity(KR)
(7) IEF US Equity	iShares 7-10 Year Treasury Bond ETF	Bond(US)
(8) GLTY LN Equity	SPDR Bloomberg UK Gilt UCITS ETF	Bond(UK)
(9) RXP5EX GR Equity	iShares eb.rexx Government Germany 5.5-10.5 UCITS ETF DE	Bond(DE)
(10) 2510 JP Equity	NEXT FUNDS Japan Bond NOMURA-BPI Exchange Traded Fund	Bond(JP)
(11) 2813 HK Equity	ChinaAMC Bloomberg Barclays China Treasury + Policy Bank Bond Index ETF	Bond(CH)
(12) 114260 KS Equity	Samsung Kodex Treasury Bond Securities ETF	Bond(KR)
(13) SHY US Equity	iShares 1-3 Year Treasury Bond ETF	Bond(US)
(14) BIL US Equity	SPDR Bloomberg 1-3 Month T-Bill ETF	Bond(US)
(15) GLD US Equity	SPDR Gold Shares	Commodity
(16) XLC US Equity	Communication Services Select Sector SPDR Fund	Equity(US)
(17) XLY US Equity	Consumer Discretionary Select Sector SPDR Fund	Equity(US)
(18) XLP US Equity	Consumer Staples Select Sector SPDR Fund	Equity(US)
(19) XLE US Equity	Energy Select Sector SPDR Fund	Equity(US)
(20) XLF US Equity	Financial Select Sector SPDR Fund	Equity(US)
(21) XLV US Equity	Health Care Select Sector SPDR Fund	Equity(US)
(22) XLI US Equity	Industrial Select Sector SPDR Fund	Equity(US)
(23) XLB US Equity	Materials Select Sector SPDR Fund	Equity(US)
(24) XLRE US Equity	The Real Estate Select Sector SPDR Fund	Equity(US)
(25) XLK US Equity	Technology Select Sector SPDR Fund	Equity(US)
(26) XLU US Equity	Utilities Select Sector SPDR Fund	Equity(US)

TABLE 4. Experiment data set list.

Experiment Data Set	ETF List	Description
ALL Country(ALL)	(1), (2), (3), (4), (5), (6), (7), (8), (9), (10), (11), (12)	"The 'ALL' dataset consists of an ETF that tracks the market indices of developed and emerging markets, a long-term bond ETF, and a commodity ETF."
Developed Market(DM)	(1), (2), (3), (4), (7), (8), (9), (10)	"The 'DM' dataset consists of an ETF that tracks the market indices of developed markets, a long-term bond ETF, and a commodity ETF."
Global Bond	(7), (13), (14), (8), (9), (10)	"The 'Global Bond' dataset is constructed by referencing the Bloomberg Global Aggregate Total Return Index and taking into account the maturity and country of the bonds."
US Sector	(16), (17), (18), (19), (20), (21), (22), (23), (24), (25), (26)	"The 'US Sector' dataset consists of 11 Sector ETFs in the United States, with reference to the Global Industry Classification Standard (GICS)."

important in real investment. We define a utility function according to the degree of risk aversion. This function can be considered as a utility function according to the investment type. The utility function [1], [41] is defined as follows.

Definition (Utility Function):

$$U(r, \sigma, k) = r - k \times \sigma \quad (13)$$

where: r : portfolio return, σ : portfolio volatility, k : Control Parameters for Risk Aversion

We define different types of investors based on the risk aversion parameter value. The larger the risk aversion parameter value, the more the investor tends to avoid risk. The

smaller the risk aversion parameter value, the less the investor tends to avoid risk [1], [41], [42]. For this purpose, we set the risk aversion control parameter to 0.2, 0.4, 0.6, and 0.8.

E. EXPERIMENT PERIOD

The Experiment Period for ALL, DM, and US Sector data sets is set as follows Table 4. In the case of the Global Bond dataset, a particular ETF was established in 2013. In other words, data on specific ETFs have been collected since 2013. With the ETF that has the shortest historical data, the Experiment Period for Global Bond are determined as Table 5.

TABLE 5. Test period for ALL, DM, US sector.

Test Number	Test Period
1	2008-02-18 ~ 2010-02-18
2	2010-02-18 ~ 2012-02-18
3	2012-02-18 ~ 2014-02-18
4	2014-02-18 ~ 2016-02-18
5	2016-02-18 ~ 2018-02-18
6	2018-02-18 ~ 2020-02-18
7	2020-02-18 ~ 2022-02-18

TABLE 6. Test period for global bond.

Test Number	Test Period
1	2013-02-18 ~ 2015-02-18
2	2015-02-18 ~ 2017-02-18
3	2017-02-18 ~ 2019-02-18
4	2019-02-18 ~ 2021-02-18
5	2021-02-18 ~ 2023-02-18

F. EXPERIMENT RESULT

our method in the proposed framework are trained using only simulation data generated by black-scholes merton [31]. For our method, all experimental data is unseen and test data.

Overall, models are evaluated as follows. Markowitz pursues high returns, but shows characteristics that are vulnerable to risk management due to high volatility and maximum drawdown. This means that Markowitz is a model in which the value of the portfolio can fluctuate rapidly. And investors using Markowitz can suffer large losses in a bear market. The risk budgeting model has low volatility and MDD, but the return is relatively very low, so there is a limit to maximizing investment returns. This characteristic may make it difficult for investors to achieve their investment goals in terms of returns. And if we consider inflation, there is a disadvantage that the real return of risk budgeting is lowered even further. Risk budgeting focuses on minimizing risk. This characteristic has a serious disadvantage that it can miss opportunities to generate returns. This characteristic makes risk budgeting unsuitable for investors who want to increase their assets in the long term. Equal weight is a methodology that gives equal weight to all assets. This is an inefficient method that does not consider the risk and return characteristics of assets. Equal weight shows intermediate performance in terms of returns and risks. Equal weight is a methodology with no particular strengths. Therefore, it is difficult to obtain the optimal results that fit the investment purpose. In addition, this method has the disadvantage of being difficult to flexibly respond to market changes. DQN and A3C are deep learning-reinforcement learning-based benchmarks. Unlike our method, these models are trained using real data. The training data is 5 years of data before each test period. A3C converge faster than DQN in the training period, but its performance in the test period is similar. This shows that there is no difference in performance according to the reinforcement learning algorithm in the test period of non-stationary financial time series data.

In addition, these models tend to learn only specific patterns in the training period and perform poorly in the test period, or perform well when the training pattern exists in the test period. In other words, it can be confirmed that general reinforcement learning models do not work well when the statistics of the training period and the statistics of the test period are different. We think that financial analysis of these unstable models would be difficult. Our method has shown that it pursues high returns while effectively managing risks to increase investment efficiency. And this helps both investors achieve their financial goals and manage risks. Our method has shown consistent performance in various market situations. This means that our method can flexibly respond to market changes.

ALL dataset is for global markets. It includes stock market indices and bond indices of major countries in the United States, Europe, and Asia. Overall, our method provides higher average returns than risk budgeting and equal weight. And our method keeps volatility and MDD significantly lower than Markowitz. Our method also achieves an average sharpe value of 0.71, which indicates efficient risk-adjusted returns. These results show that our method outperforms equal weight and Markowitz, and competes closely with risk budgeting. And our method shows consistent performance indicators in various market conditions over a 14-year testing period. In the period 1 (global financial crisis), our method record the highest return compared to other benchmarks, and record low volatility and low MDD. In addition, our method show excellent risk-adjusted return and great advantages in terms of risk management compared to Markowitz. This can be considered a model that provides investors with excellent investment performance. In the period 5 (US stock bull market), Markowitz record high returns but had high volatility and MDD. In other words, the risk-adjusted return is poor. Our method show excellent performance in risk-adjusted efficiency with stable returns and low volatility. In the case of risk budgeting, the risk-adjusted return is poor despite the low volatility due to the excessively low returns. In the periods 6 (US-China trade war) and 7 (COVID-19), Markowitz record high returns but have high volatility and MDD. Our method provid high returns while maintaining low volatility and MDD at similar levels to the risk budgeting model. And when comparing DQN, A3C and our method, DQN, A3C are confirmed to have lower profitability and higher volatility than our method. And the absolute value of MDD is large. A large absolute value of MDD means that the model does not work well for unseen data. In other words, we think that the test period of All Dataset contains data with various patterns other than the patterns learned from the training data, so the DQN, A3C models do not work well. In other experimental periods (periods 2, 3, and 4), it do not show significantly better performance than the benchmarks, but it achieve stable returns. And it show excellent risk management with low volatility and MDD.

The DM dataset is for global developed markets. The dataset includes stock market indices and bond indices of

TABLE 7. ALL experiment result - traditional method.

Period	MV				RB				EW			
	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe
(1)	-8.97%	59.70%	-64.95%	-0.05	0.62%	4.74%	-8.27%	0.13	1.93%	11.45%	-22.69%	0.16
(2)	15.27%	20.78%	-24.14%	0.73	4.45%	2.77%	-2.41%	1.6	6.77%	6.92%	-9.37%	0.97
(3)	8.97%	13.46%	-14.72%	0.66	0.90%	2.42%	-4.86%	0.37	2.26%	5.34%	-6.99%	0.42
(4)	4.67%	19.95%	-27.55%	0.23	2.59%	2.09%	-4.66%	1.23	0.76%	5.35%	-12.13%	0.14
(5)	17.38%	17.96%	-15.26%	0.96	1.12%	1.85%	-3.49%	0.6	6.33%	4.36%	-5.04%	1.45
(6)	9.26%	14.90%	-21.72%	0.62	2.68%	2.16%	-3.98%	1.24	3.18%	4.62%	-7.54%	0.68
(7)	18.86%	24.81%	-32.26%	0.76	1.80%	3.48%	-7.20%	0.51	7.24%	8.14%	-15.04%	0.88
Mean	9.35%	24.51%	-64.95%	0.38	2.02%	2.79%	-8.27%	0.72	4.07%	6.60%	-22.69%	0.61

TABLE 8. ALL experiment result - deep learning.

Period	DQN				A3C				Model			
	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharp	R	Sigma	MDD	Sharp
(1)	0.24%	9.31%	-18.92%	0.03	1.57%	10.71%	-19.46%	0.15	2.68%	7.18%	-11.74%	0.36
(2)	3.40%	8.16%	-10.83%	0.42	3.70%	8.11%	-10.88%	0.46	4.06%	4.92%	-4.16%	0.82
(3)	1.66%	6.46%	-8.06%	0.26	2.74%	6.12%	-7.39%	0.45	1.49%	4.39%	-5.81%	0.34
(4)	-2.14%	5.00%	-9.91%	-0.43	-1.43.27%	5.94%	-11.05%	-0.24	0.25%	3.89%	-7.52%	0.06
(5)	8.00%	5.61%	-6.12%	1.43	8.79%	6.86%	-5.98%	1.28	4.38%	3.62%	-4.72%	1.21
(6)	1.70%	5.28%	-9.89%	0.32	0.96%	5.86%	-11.07%	0.16	3.96%	3.65%	-4.74%	1.08
(7)	1.40%	10.42%	-18.06%	0.13	2.76%	11.43%	-19.14%	0.24	7.85%	7.09%	-11.88%	1.1
Mean	2.04%	7.18%	-18.92%	0.28	2.73%	7.86%	-19.46%	0.35	3.52%	4.96%	-11.88%	0.71

TABLE 9. DM experiment result - traditional method.

Period	MV				RB				EW			
	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe
(1)	6.60%	14.27%	-18.33%	0.46	0.23%	4.25%	-9.81%	0.05	-2.01%	10.98%	-23.69%	-0.18
(2)	-0.89%	18.39%	-23.68%	-0.04	4.53%	3.06%	-3.39%	1.48	5.12%	7.01%	-6.86%	0.73
(3)	27.21%	18.89%	-20.44%	1.44	3.08%	2.77%	-5.39%	1.11	8.96%	5.24%	-6.81%	1.70
(4)	-4.56%	17.52%	-23.22%	-0.26	1.89%	2.83%	-4.95%	0.66	2.34%	5.89%	-8.86%	0.39
(5)	9.17%	11.29%	-9.57%	0.81	3.70%	2.41%	-2.25%	1.53	9.397%	4.88%	-2.83%	2.04
(6)	3.65%	12.41%	-12.89%	0.29	4.10%	2.67%	-3.10%	1.53	5.23%	4.76%	-8.07%	1.09
(7)	-0.52%	31.80%	-34.30%	-0.01	-0.94%	5.11%	-8.50%	-0.18	5.00%	11.22%	-16.81%	0.44
Mean	5.81%	17.80%	-34.30%	0.32	2.37%	3.30%	-9.81%	0.71	4.94%	7.14%	-23.69%	0.69

TABLE 10. DM experiment result - deep learning.

Period	DQN				A3C				Model			
	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe
(1)	0.28%	10.13%	-20.55%	0.03	-1.32%	10.91%	-22.60%	-0.12	4.26%	6.89%	-12.02%	0.61
(2)	6.78%	8.19%	-9.20%	0.83	5.54%	8.84%	-10.70%	0.63	5.24%	6.65%	-6.56%	0.78
(3)	3.69%	7.72%	-9.89%	0.48	3.02%	6.77%	-6.92%	0.45	10.07%	5.62%	-7.15%	1.79
(4)	-0.91%	5.56%	-8.30%	-0.16	0.63%	5.87%	-8.65%	0.11	4.22%	5.59%	-8.46%	0.75
(5)	9.12%	7.56%	-6.85%	1.21	6.00%	6.07%	-7.81%	0.99	10.58%	4.39%	-2.95%	2.41
(6)	1.52%	5.98%	-12.38%	0.25	1.64%	5.59%	-12.43%	0.29	5.30%	4.91%	-7.25%	1.07
(7)	4.40%	10.27%	-17.29%	0.43	2.39%	10.59%	-18.76%	0.23	5.00%	12.75%	-19.33%	0.39
Mean	3.55%	7.92%	-20.55%	0.45	2.56%	7.81%	-22.60%	0.33	6.38%	6.69%	-19.33%	0.95

developed markets in the United States, Europe, and Asia. Overall, our method showed similar results to ALL Experiment. Our method showed an excellent risk-return balance by providing the highest average return while maintaining low volatility and MDD. Markowitz provides high returns, but has high volatility and MDD, which is a big risk. Risk budgeting showed the lowest volatility and MDD, but has the disadvantage of too low returns. As with the ALL dataset, the test periods of the DM dataset also contains various data patterns other than the training data patterns. This means that DQN and A3C have difficulty operating in these test periods. As a result, DQN and A3C show poor results in all models. On the other hand, our method has a similar level of risk management ability to risk budgeting while providing higher returns. When comparing the risk-return ratio through the Sharpe index, our method showed the highest efficient investment performance with 1.10. This

means that our method shows better performance than risk budgeting and equal weight. In Period 1 (global financial crisis), our method has a higher Sharpe ratio than other benchmarks. This means that our method has a high risk-adjusted return. Even though the stock market volatility is high in that period, our method control the volatility well compared to other methodologies. This characteristic is also observed in Period 6 (US-China trade war) and Period 7 (COVID-19). In addition, in Period 5 (US stock bull market), our method record the highest return and showed efficient investment performance due to low volatility. Our method is superior to other models in both profitability and stability. Our method show high return and appropriate volatility in the DM dataset, and its risk-adjusted efficiency is superior to other models. In addition, it show consistent performance in market conditions and provide stable returns even in periods of high volatility.

TABLE 11. US sector experiment result - traditional method.

Period	MV				RB				EW			
	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe
(1)	-9.33%	35.92%	-50.40%	-0.25	-5.49%	29.40%	-49.40%	-0.18	-5.04%	32.29%	-52.54%	-0.15
(2)	0.86%	23.21%	-32.31%	0.03	7.25%	18.75%	-16.65%	0.38	7.02%	20.26%	-19.12%	0.34
(3)	9.16%	14.16%	-13.35%	0.64	14.23%	10.94%	-7.04%	1.30	14.46%	11.62%	-8.76%	1.24
(4)	-9.97%	17.02%	-27.62%	-0.58	-0.71%	13.10%	-12.68%	-0.05	-1.36%	13.42%	-13.47%	-0.10
(5)	31.64%	13.88%	-7.06%	2.27	16.03%	7.89%	-5.57%	2.03	17.56%	8.35%	-5.51%	2.10
(6)	6.07%	15.66%	-19.05%	0.38	8.36%	11.58%	-17.34%	0.72	8.00%	12.80%	-19.37%	0.625
(7)	11.33%	33.36%	-24.94%	0.33	20.04%	25.35%	-31.77%	0.79	21.90%	26.23%	-32.24%	0.83
Mean	5.68%	21.89%	-50.40%	0.25	8.53%	16.72%	-49.40%	0.51	8.93%	17.85%	-52.54%	0.50

TABLE 12. US sector experiment result - deep learning.

Period	DQN				A3C				Model			
	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe
(1)	-1.57%	18.77%	-31.94%	-0.08	-1.32%	24.63%	-42.65%	-0.05	-3.43%	33.89%	-50.62%	-0.10
(2)	5.17%	13.99%	-14.14%	0.37	5.39%	14.92%	-14.40%	0.36	10.20%	20.48%	-20.48%	0.49
(3)	11.47%	8.92%	-6.01%	1.29	12.17%	9.10%	-5.98%	1.34	15.85%	12.84%	-10.38%	1.23
(4)	1.77%	10.47%	-10.70%	0.17	2.50%	11.77%	-12.37%	0.21	-0.43%	14.53%	-15.17%	-0.02
(5)	12.43%	8.31%	-8.85%	1.50	13.98%	8.41%	-8.38%	1.66	20.47%	9.62%	-5.46%	2.12
(6)	11.02%	12.52%	-19.34%	0.88	11.17%	11.67%	-18.45%	0.96	7.28%	14.97%	-21.96%	0.48
(7)	12.82%	25.05%	-34.61%	0.51	10.72%	25.07%	-33.93%	0.43	19.83%	27.36%	-39.09%	0.72
Mean	7.59%	14.00%	-34.61%	0.54	7.80%	15.08%	-42.65%	0.52	9.97%	19.10%	-50.62%	0.52

The US sector dataset is consisting of 11 US stock sectors. This dataset represents the US stock market. Overall, our method has shown excellent ability to minimize losses during financial crises or bear market. This is a great advantage for investors in terms of risk management. Over multiple test periods, our method has consistently achieved high returns. In particular, our method has outperformed the benchmarks in bull markets and recovery periods. Based on the Sharpe ratio, our method has recorded high returns relative to risk in most periods. In addition, our method has the advantage of maintaining relatively stable performance even in volatile markets, which can provide confidence to investors. In period 1 (global financial crisis), our method show less decline in returns compared to traditional benchmarks. The US Sector dataset is the data that constitutes the US Stock index. These data generally have an upward pattern. This makes the training data pattern and the test data pattern similar. This makes DQN and A3C show very good performance compared to other models. These results demonstrate the non-stationary characteristics of financial time series that we claim, and show that our methodology is suitable for financial time series analysis. In periods 2, 3 (financial market recovery), the US implement economic stimulus and easing monetary policies. Our method show the highest returns compared to other benchmarks. And our method have high Sharpe ratio on average. This means that our method actively generated returns during the market recovery while effectively managing risks. In period 5 (US stock bull market), Our method show the second highest returns after Markowitz. And our method stably control volatility, resulting in a high Sharpe ratio of 2.12. This means that our method pursued high returns while effectively managing risks. Our method has consistently outperformed the benchmarks in various environments of the US stock market. In addition, our method has shown that it minimizes losses and protects assets during periods of market decline

or increased volatility. Considering these aspects, it can be said that our method has shown consistent performance. This means that it can provide better investment results to investors.

The Global Bond experiment dataset is intended for experimenting with asset allocation for bonds with varying countries and durations.(Table 6) The primary objective of the Global Bond experimental dataset is to effectively control risk in situations where returns are generally low. Similar to the ALL and DM datasets, Markowitz tends to not effectively manage risk. In cases where risk exists in the bond dataset, there is a tendency to increase the weight of SHY US Equity and BIL US Equity to reduce risk, but instead, there is a higher allocation to long-term bonds, leading to higher volatility. Markowitz demonstrates less favorable experimental results compared to our method. Risk budgeting shows a tendency to manage risk. For the Global Bond experimental dataset, it outperforms our method.

The calmar ratio is a statistic that uses the maximum loss and return of the fund. This is an indicator that customers must look at when subscribing to a fund. In other words, it is an indicator that reflects how much maximum loss customers can bear. We calculate the calmar ratio using the average return and maximum MDD in each dataset. In the case of Markowitz, volatility is large, so the maximum drawdown, MDD, tends to be large. Therefore, no matter how high the return is, the calmar ratio is low. In the case of risk budgeting, volatility is very small, so there is a tendency for the MDD to be small, but the return is excessively low. In the case of equal weight, volatility is lower than Markowitz, but the volatility tends to be higher than other benchmarks. Therefore, the calmar ratio tends to be low. Our method has a significantly lower MDD than benchmarks excluding risk budgeting. And our method has the characteristic of high returns except for Markowitz. Therefore, our method tends to have a higher Calmar ratio than other benchmarks. If we

TABLE 13. Global bond experiment result - traditional method.

Period	MV				RB				EW			
	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe
(1)	2.26%	4.44%	-5.72%	0.50	0.51%	0.74%	-0.82%	0.68	0.74%	2.65%	-4.90%	0.27
(2)	1.85%	6.22%	-6.98%	0.29	0.10%	1.03%	-1.28%	0.09	0.05%	3.35%	-4.24%	0.01
(3)	-0.16%	4.28%	-5.21%	-0.03	-0.23%	1%	-1.86%	-0.23	-0.23%	2.44%	-3.72%	-0.09
(4)	0.31%	6.59%	-9.15%	0.04	0.85%	0.91%	-0.85%	0.93	1.69%	3.70%	-3.87%	0.45
(5)	-8.91%	5.81%	-17.21%	-1.53	-0.96%	0.77%	-2.14%	-1.24	-6.77%	5.27%	-16.48%	-1.28
Mean	-0.93%	5.47%	-17.21%	-0.17	0.05%	0.89%	-2.14%	0.06	-0.90%	3.48%	-16.48%	-0.25

TABLE 14. Global bond experiment result - deep learning.

Period	DQN				A3C				Model			
	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe	R	Sigma	MDD	Sharpe
(1)	0.53%	3.68%	-4.45%	0.14	0.26%	2.88%	-3.21%	0.09	1.28%	2.57%	-3.73%	0.49
(2)	-1.24%	3.62%	-4.82%	-0.34	-1.42%	3.68%	-5.53%	-0.39	0.03%	3.11%	-3.01%	0.00
(3)	1.56%	3.62%	-5.27%	0.43	0.71%	2.92%	-4.44%	0.24	-1.25%	2.03%	-4.45%	-0.61
(4)	2.79%	5.26%	-9.52%	0.53	2.71%	3.73%	-5.45%	0.73	2.69%	4.69%	-5.45%	0.57
(5)	-11.71%	7.67%	-27.03%	-1.53	-10.03%	5.71%	-22.09%	-1.76	-3.66%	3.27%	-9.23%	-1.11
Mean	-1.61%	4.77%	-27.03%	-0.34	-1.55%	3.78%	-22.09%	-0.41	-0.18%	3.13%	-9.23%	-0.05

look at the experimental results from an actual investment perspective, we can see that our method outperforms the benchmark. When the maximum loss occurs, actual fund customers tend to have a hard time recovering their losses. Therefore, our method, which has low MDD and high returns, can be seen as a good model for managing funds. The experimental results for the calmar ratio can be found in Table 15.

We apply the perspective of real investment to evaluate the method.(Table 16, 17, 18, 19) For this, we use utility functions. We use the utility function of Risk Aversion($K = 0.2, 0.4, 0.6, 0.8$). In other words, we interpret the returns and volatility of the models from the perspective of investment risk and use this to confirm that our method is better than other benchmarks. The returns and volatility required to calculate the utility function are the average returns and average volatility of the model for each experimental data set.

TABLE 15. Calmar ratio experiment result.

	MV	RB	EW	DQN	A3C	Model
ALL	0.33	0.41	0.36	0.11	0.14	0.49
DM	0.29	0.44	0.46	0.17	0.11	0.70
Global Bond	-0.10	0.039	-0.13	0.22	0.18	-0.035
US Sector	0.22	0.42	0.41	-0.06	-0.07	0.43

TABLE 16. Utility function for ALL data set - experiment result.

Risk Aversion (K)	MV	RB	EW	DQN	A3C	Model
K=0.2	4.45%	1.47%	2.75%	0.60%	1.15%	2.53%
K=0.4	-0.45%	0.90%	1.43%	-0.83%	-0.42%	1.54%
K=0.6	-5.36%	0.35%	0.11%	-2.27%	-1.99%	0.54%
K=0.8	-10.26%	-0.21%	-1.21%	-3.70%	-3.56%	-0.45%

The Markowitz model generally shows experimental results showing very high volatility compared to the return. As a result, the utility function value of Markowitz is very bad compared to the utility function value of other models in all data sets. In other words, the Markowitz model is the worst

model in satisfying the risk type of investors in various data set. In other words, it is dangerous to use Markowitz in real investment.

The RB model controls volatility. Therefore, the RB model shows very low volatility in all data sets compared to other models. Since this RB is a model that controls risk, the asset weighting tends to be concentrated in bonds. Due to these characteristics of the model, the RB model is considered a model that is simply suitable for risk-averse investors. In other words, investors who are not risk-averse investors should not use this model.

The EW model assigns equal weights to all assets. This model produces higher utility function values than the Markowitz and RB models as it approaches risk-seeking type. However, the utility function values do not outperform our method.

TABLE 17. Utility function for DM data set - experiment result.

Risk Aversion (K)	MV	RB	EW	DQN	A3C	Model
K=0.2	2.25%	1.71%	3.43%	1.97%	1.00%	5.04%
K=0.4	-1.31%	1.05%	2.01%	0.39%	-0.57%	3.71%
K=0.6	-4.87%	0.39%	0.58%	-1.20%	-2.13%	2.37%
K=0.8	-8.43%	-0.27%	-0.85%	-2.78%	-3.69%	1.03%

TABLE 18. Utility function for global bond data set - experiment result.

Risk Aversion (K)	MV	RB	EW	DQN	A3C	Model
K=0.2	-2.02%	-0.12%	-1.60%	-2.57%	-2.31%	-0.81%
K=0.4	-3.11%	-0.30%	-2.29%	-3.52%	-3.07%	-1.43%
K=0.6	-4.21%	-0.48%	-2.98%	-4.48%	-3.82%	-2.05%
K=0.8	-5.36%	-0.66%	-3.68%	-5.43%	-4.58%	-2.68%

DQN and A3C are deep learning-reinforcement learning-based models. These models have the characteristic that when the pattern of the training data appears in the test section, the model performance is good, and when it does not, the model performance is bad. In the case of ALL, DM, and Global Bond datasets, since they contain data with various

TABLE 19. Utility function for US sector data set - experiment result.

Risk Aversion (K)	MV	RB	EW	DQN	A3C	Model
K=0.2	1.30%	5.19%	5.36%	4.79%	4.79%	6.15%
K=0.4	-3.07%	1.84%	1.79%	1.99%	1.77%	2.33%
K=0.6	-7.45%	-1.50%	-1.78%	-0.82%	-1.25%	-1.49%
K=0.8	-11.83%	-4.84%	-5.35%	-3.62%	-4.26%	-5.31%

patterns, the DQN and A3C models performed worse than other models. Accordingly, the utility function results are also worse than our method.

Our method outperforms the utility functions of other models on the dataset except for the Global Bond dataset. That is, the portfolio managed by our method can be interpreted as generating the best value in various utility functions in the return/risk plane. And this is shown in the ALL, DM, and US Sector datasets. This shows that our method is the best model for all risk types when compared to traditional asset allocation models in the global financial market.

VI. CONCLUSION

We avoid overfitting of the model by utilizing Monte Carlo simulation data. Additionally, we consider the statistical structure of non-stationary financial markets in the model training. Furthermore, the state of the reinforcement learning agent takes into account both the characteristics of the financial market and the performance of the managed portfolio simultaneously. Our method demonstrates superior performance over benchmarks in various experiment datasets with different features, including ALL, DM, Global Bond, and US Sector. In other words, this shows that financial artificial intelligence models, facing challenges in model training due to data scarcity, can outperform conventional financial methodologies. In fact, our method has advantages that make it easily applicable in real portfolio management. We plan to develop our methodology as a hierarchical multi-agent system, which is expected to replace Strategic Asset Allocation (SAA) and Tactical Asset Allocation (TAA) commonly used in asset allocation. This is our future work.

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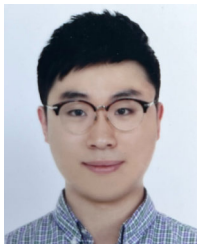


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